

$$P(\text{buy} = y \mid \hat{x}) = P(\hat{x} \mid \text{buy} = \text{yes}) P(\text{buy} = \text{yes})$$

$$P(\text{age} \leq 30 \mid \text{yes}) \quad P(\text{student} = \text{yes} \mid \text{yes}) \quad P(\text{buy} = \text{yes} \mid \text{yes})$$

$$\frac{2}{9} \quad \frac{6}{9} \quad \frac{9}{14}$$

$$P(y \mid \hat{x}) = \frac{2}{9} \times \frac{6}{9} \times \frac{9}{14}$$